

Engineering Mechanics: ME101

Lecture 5: Statics



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Contents

- ✓ Structures: Difference between trusses,
- ✓ Frames and machines,
- ✓ Truss analysis methods – Method of joints,
- ✓ Introduction to Method of sections

Structural Analysis

Engineering Structure



An engineering structure is any connected system of members built to support or transfer forces and to safely withstand the loads applied to it.

Different Structure used Engineering and their name and brief concept

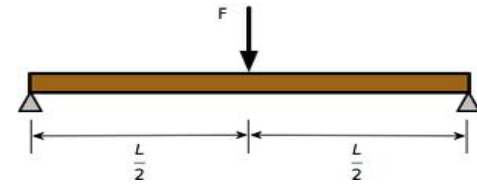
❖ **Beams:** A beam may be defined as an element in which one dimension is much greater than the other two and the applied loads are usually normal to the main axis of the element.



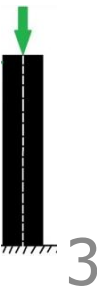
✓ **Cantilever:** Supported at one end only with a fixed connection.

✓ **Simply supported:** Supported vertically at each end; horizontally on only one to withstand friction, and able to rotate at the supports.

✓ **Continuous:** Supported by three or more supports.



❖ **Columns:** Columns are elements that carry only axial compression force compression - or both axial force and bending (which is technically called a beam-column but practically, just a column).



Different Structure used Engineering (*contd.*)

- ❖ **Truss:** The joints are pin type joints and the members are free to rotate about the pin. As such, a truss cannot transfer moments and members are subjected to only axial forces (tensile and compression).

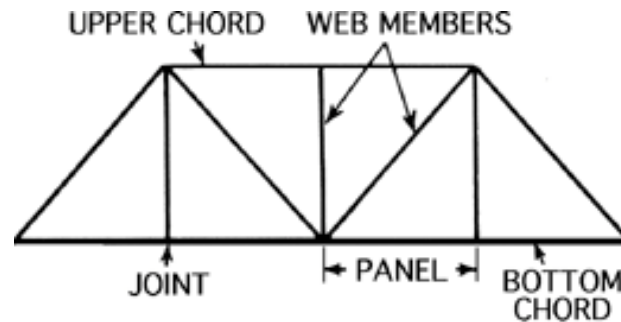


Figure 1

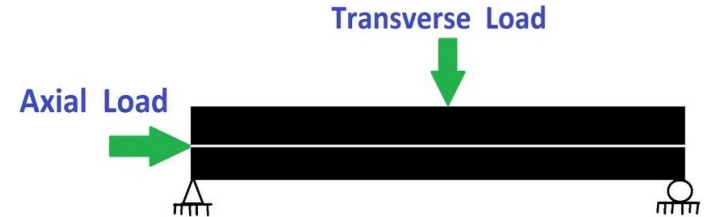


Figure 2

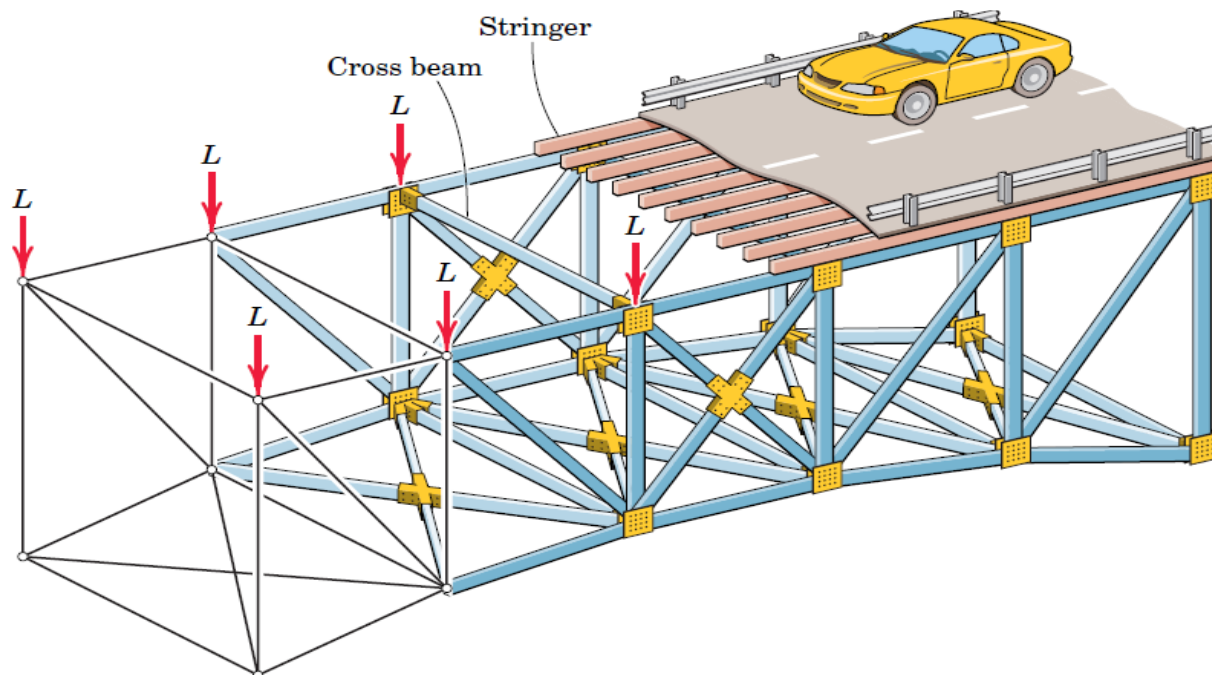
- ❖ **Frames:** On the other hand, members of frames are connected rigidly at joints by means of welding and bolting. Therefore the joints of frames can transfer moments in addition to the axial loads.
- **Trusses and frames are both composed of triangular members connected at joints.**

What is a STRUCTURE & it's types

- ❖ Structures can be classified according to their **geometrical shape and the loads acting on them.**
- ❑ If a slender structural element (cross-sectional dimensions much smaller than its length) that is loaded solely in the axial direction (tension or compression) then it is a **bar or a rod.**
- ❑ If the same geometrical object is subjected to a load perpendicular to its axis then it is a **beam.**
- ❑ A curved beam is usually designated as an **arch.**
- ❑ Structures consisting of **inclined, rigidly joined beams** are called **frames.**

Structure based on application of loads

- ❑ A plane structure with a thickness much smaller than its characteristic in-plane length is called a **disk** if it is solely loaded in its plane, e.g., by in-plane forces.
- ❑ If the same geometrical structure is loaded perpendicularly to its mid-plane it is called a **plate**.
- ❑ If such a structure is curved and loads apply then it is a **shell**.



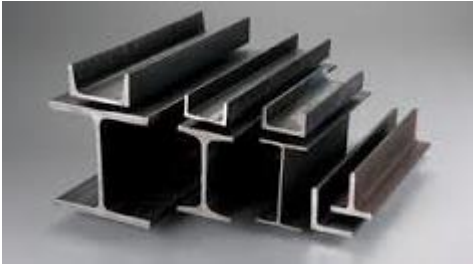
Truss

- Members of a truss are slender and not capable of supporting large lateral loads. Loads must be applied at the joints.
- ❑ A **truss** is a structure composed of slender members that are connected at their ends by joints. The truss is one of the most important structures in engineering applications.
- ✓ Bridges, roof supports, derricks, and other such structures are common examples of trusses.

TRUSS

- ✓ Structural members commonly used are I-beams, channels, angles, bars, and special shapes which are fastened together at their ends by riveted connections, or large bolts or pins.
- ✓ When the members of the truss lie essentially in a single plane, the truss is called a **plane truss**.

Types of TRUSSES



I-Beam



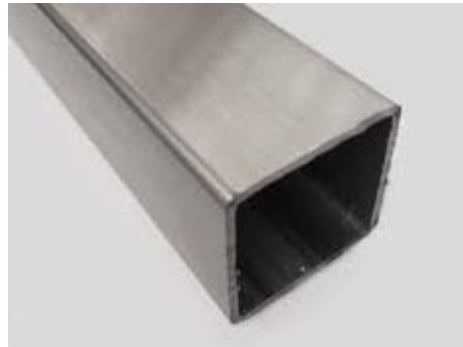
Channel



Angle



Bar



Box



wire

Definition of a Truss

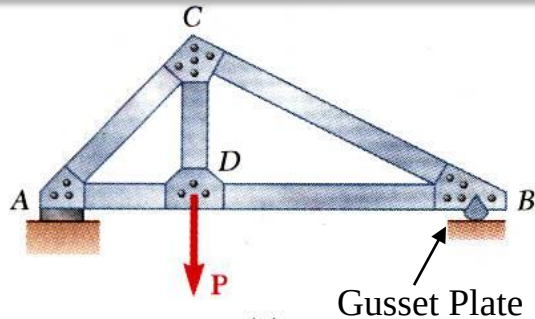


Figure 1

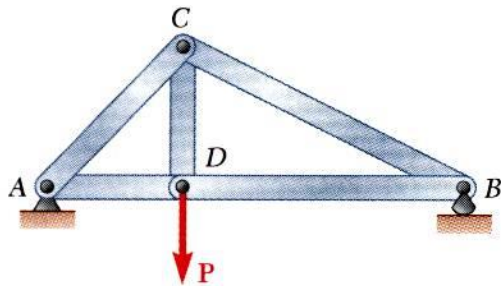


Figure 2

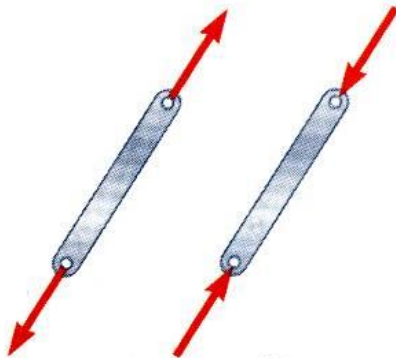
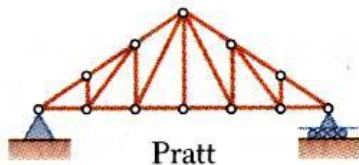


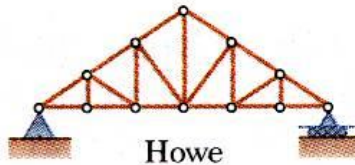
Figure 3

1. A truss consists of straight members connected at joints. **No member is continuous through a joint.**
2. **Most structures** are made of several trusses joined together to form a **space framework**. Each truss carries loads which act in its plane and may be treated as a **two-dimensional structure**.
3. **Bolted or welded connections** are assumed to be pinned together. Forces acting at the member ends reduce **to a single force and no couple**. Only *two-force members* are considered.
4. When forces tend to **pull the member apart**, it is in *tension*. **When the forces tend to compress the member**, it is in *compression*.

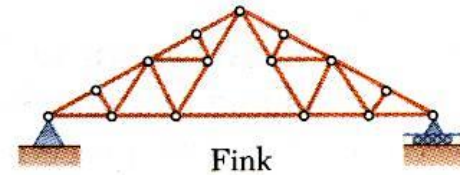
Types of Truss



Pratt

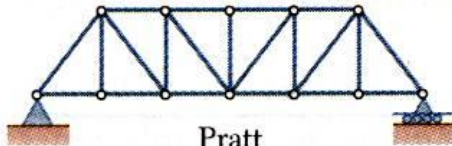


Howe

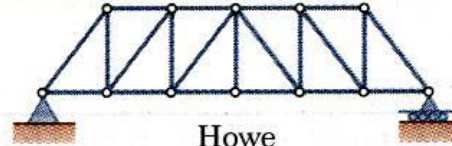


Fink

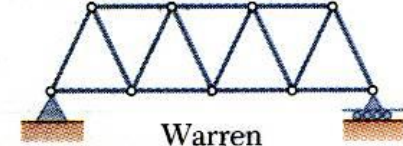
Typical Roof Trusses



Pratt



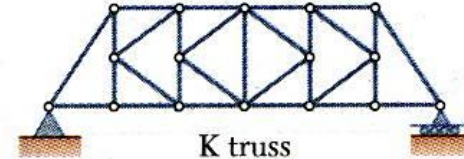
Howe



Warren

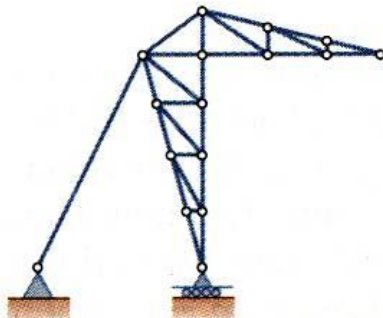


Baltimore

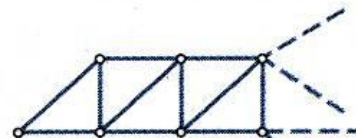


K truss

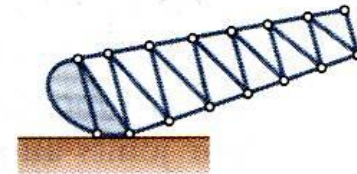
Typical Bridge Trusses



Stadium



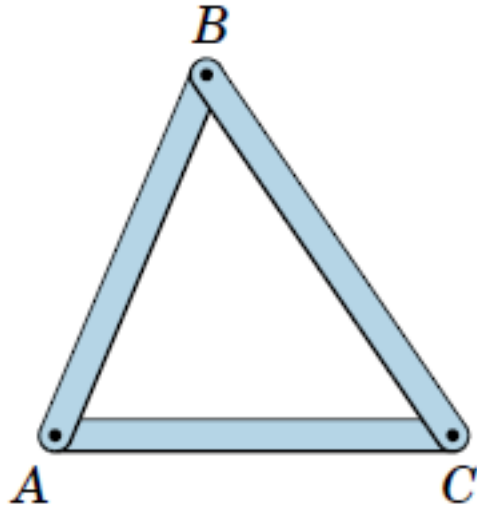
Cantilever portion
of a truss



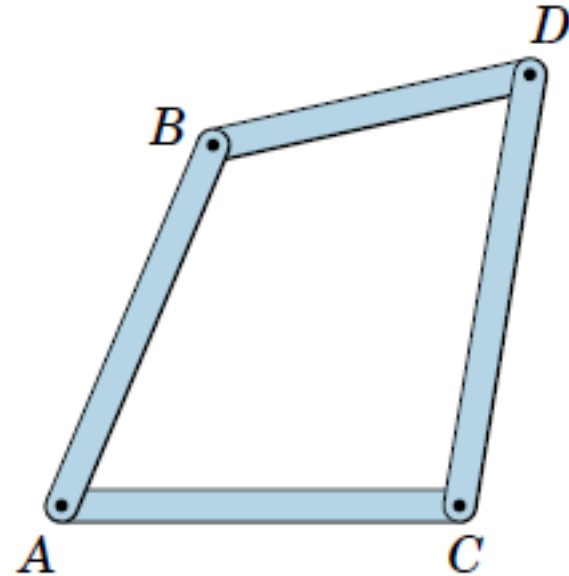
Bascule

Other Types of Trusses

Rigid and Non-Rigid Structure



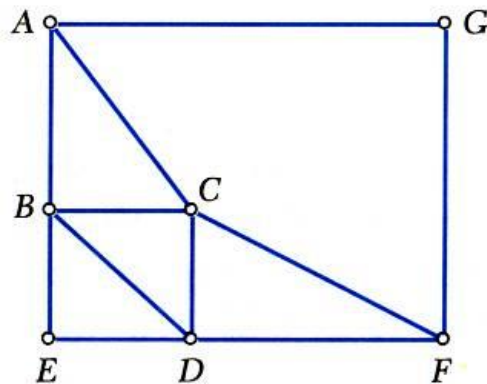
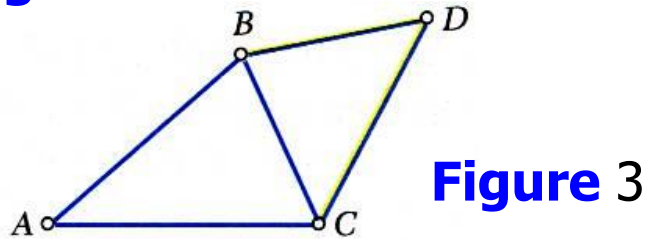
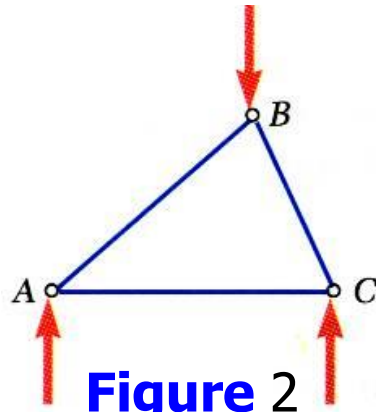
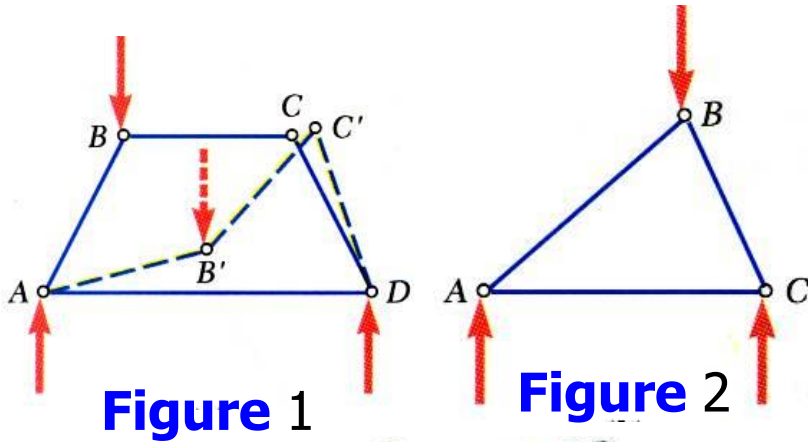
(a)



(b)

- ❑ The basic element of a plane **truss is the triangle**. Three bars joined by **pins at their ends**, Fig. *a*, constitute a rigid frame. The term **rigid** is used to mean non-collapsible and also to mean that deformation of the members due to induced internal strains is negligible.
- ❑ On the other hand, four or more bars pin-jointed to form a polygon of as many sides constitute a **nonrigid** frame.

Simple Trusses



- A *rigid truss* will not collapse under the application of a load.

- A *simple truss* is constructed by successively adding two members and one connection to the basic triangular truss.

Figure 4

Space Trusses

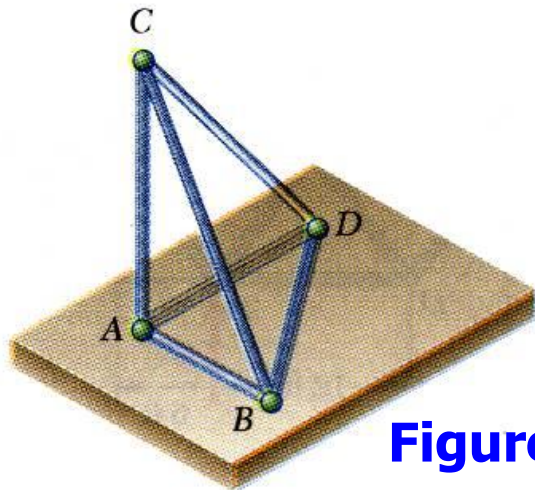


Figure 1

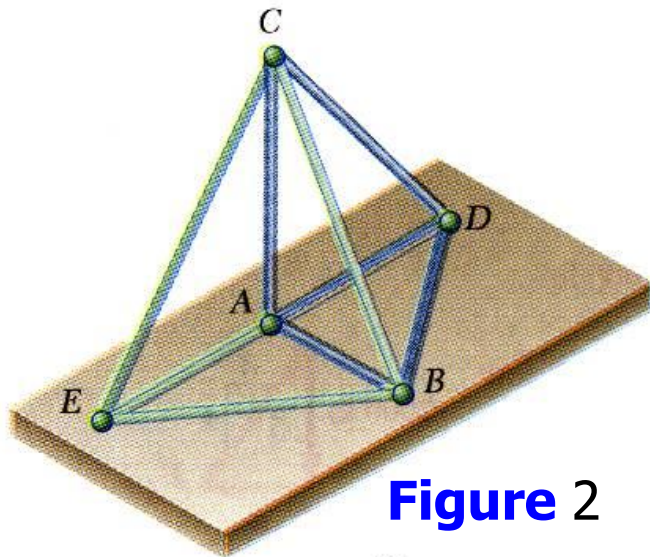


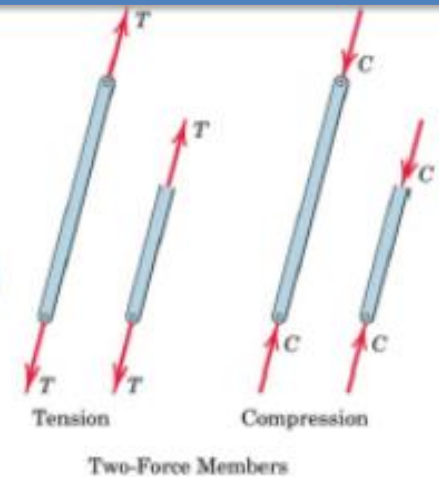
Figure 2

1. An *elementary space truss* consists of 6 members connected at 4 joints to form a tetrahedron.
2. A *simple space truss* is formed and can be extended when 3 new members and 1 joint are added at the same time.
3. In a simple space truss, $m = 3n - 6$ where m is the number of members and n is the number of joints.
4. **Conditions of equilibrium** for the joints provide $3n$ equations. For a simple truss, $3n = m + 6$ and the equations can be solved for m member forces and 6 support reactions.
5. Equilibrium for the **entire truss provides 6 additional equations** which are not independent of the joint equations.

Structural analysis of Plane Truss

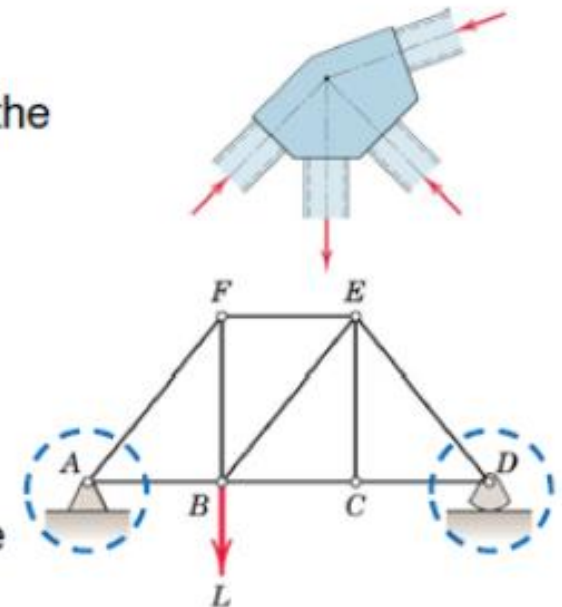
❖ 1 Basic Assumptions in Truss Analysis

- All members are two-force members.
- Weight of the members is small compared with the force it supports (weight may be considered at joints)
- No effect of bending on members even if weight is considered
- External forces are applied at the pin connections
- Pin Joint if the member centerlines are concurrent at the joint



❖ 2 Common Practice in Large Trusses

- Roller/Rocker at one end. Why?
 - to accommodate deformations due to temperature changes and applied loads.
 - otherwise it will be a statically indeterminate truss



Structural Analysis of Plane Truss by Method of Joints

Truss Analysis: Method of Joints

- Finding forces in members

Method of Joints: Conditions of equilibrium are satisfied for the forces at each joint

- **Equilibrium of concurrent forces at each joint**
- only two independent equilibrium equations are involved

Steps of Analysis

1. Draw Free Body Diagram of Truss
2. Determine external reactions by applying equilibrium equations to the whole truss
3. Perform the force analysis of the remainder of the truss by Method of Joints

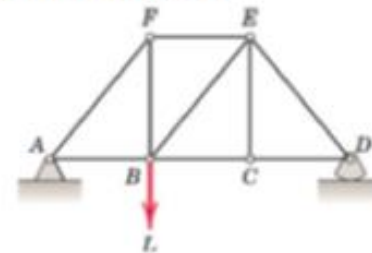


Figure 1

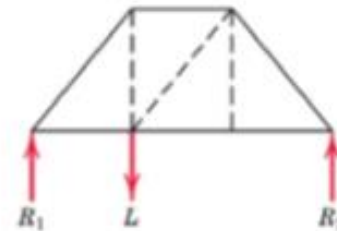


Figure 2

Truss Analysis by 'Method of Joint'

- Start with any joint where at least one known load exists and where not more than two unknown forces are present.

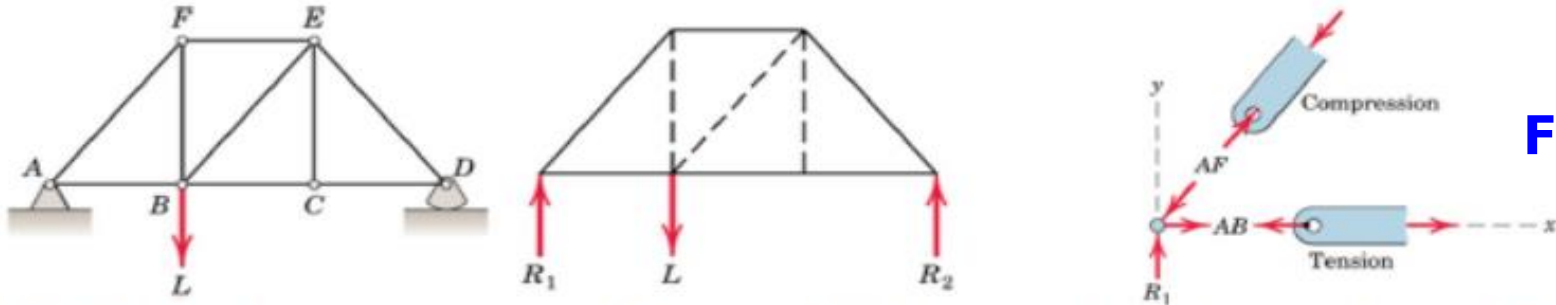


Figure 1

FBD of Joint A and members AB and AF: Magnitude of forces denoted as AB & AF

- Tension indicated by an arrow away from the pin
- Compression indicated by an arrow toward the pin

Magnitude of AF from $\sum F_y = 0$

Magnitude of AB from $\sum F_x = 0$

Analyze joints F, B, C, E, & D in that order to complete the analysis

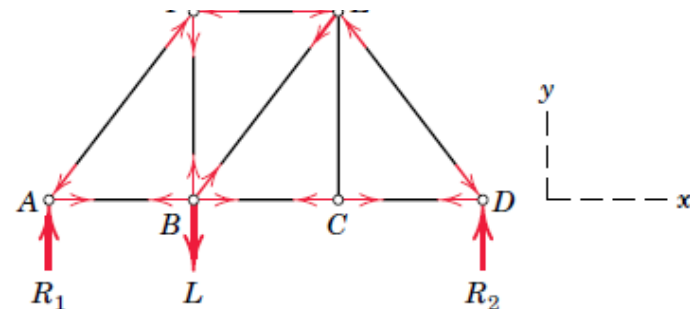


Figure 2

Analysis of Trusses by the Method of Joints

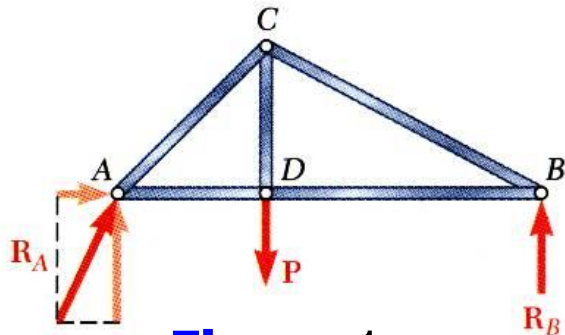


Figure 1

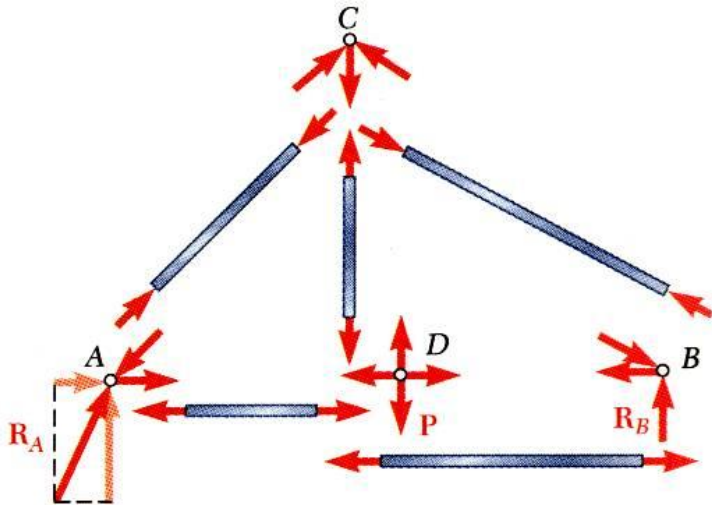


Figure 2

1. Dismember the truss and create a free body diagram for each member and pin.
2. Conditions for equilibrium for the entire truss can be used to solve for 3 support reactions.
3. The two forces exerted on each member are equal, have the same line of action, and opposite sense.
4. Forces exerted by a member on the pins or joints at its ends are directed along the member and equal and opposite.
5. Conditions of equilibrium are used to solve for 2 unknown forces at each pin (or joint), giving a total of $2n$ solutions, where n =number of joints. Forces are found by solving for unknown forces while moving from joint to joint sequentially.

Free-Body diagrams of a generalized truss

Method of Joints

Figure 1

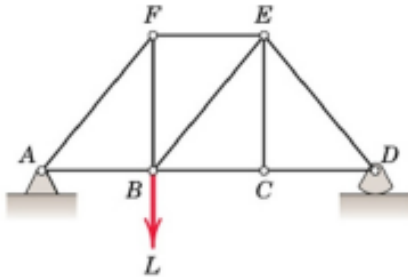
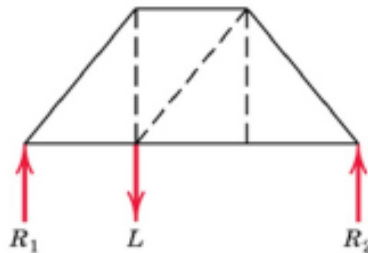
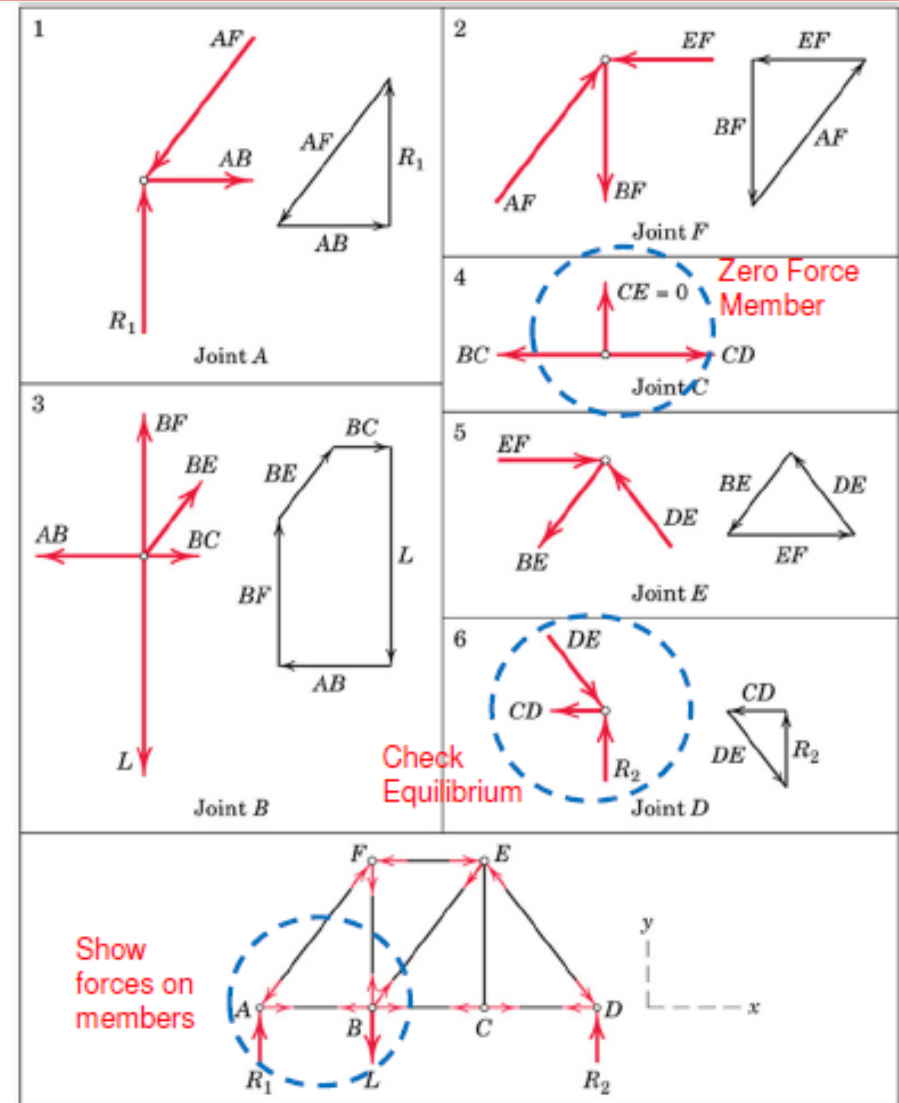


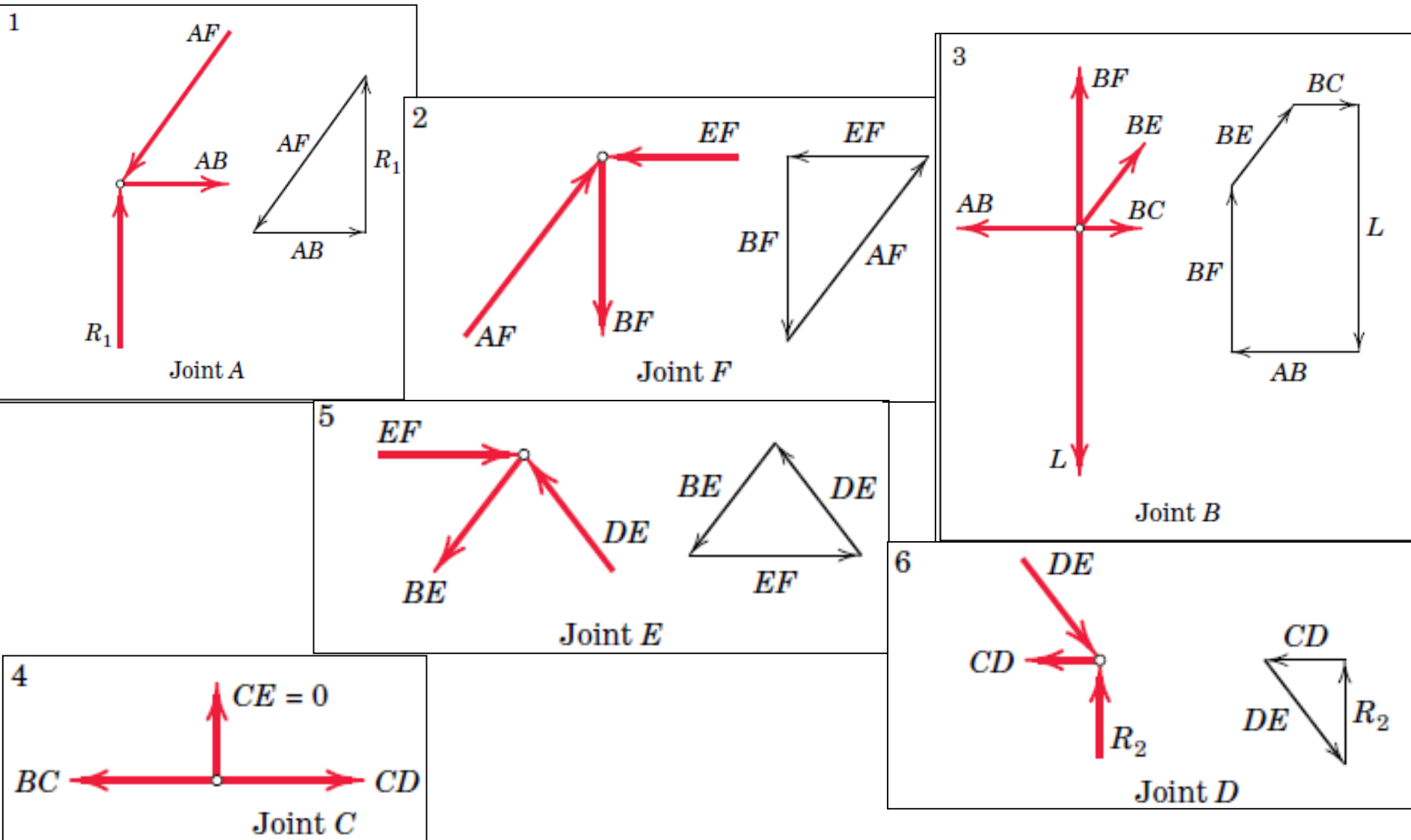
Figure 2



- Negative force if assumed sense is incorrect



Free-body diagrams of each joint



Statically Determinate Trusses

- ❖ A structure is called **statically determinate**, if the support reactions can be calculated from the three equilibrium conditions

When more number of members/supports are present than are needed to prevent collapse/stability

1. → Statically Indeterminate Truss

- cannot be analyzed using equations of equilibrium alone!
- additional members or supports which are not necessary for maintaining the equilibrium configuration → **Redundant**

2. Internal and External Redundancy

Extra Supports than required → External Redundancy

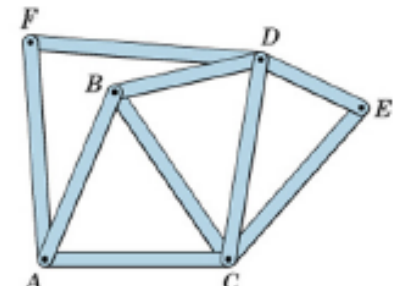
- Degree of indeterminacy from available equilibrium equations

3. Extra Members than required → Internal Redundancy

(truss must be removed from the supports to calculate internal redundancy)

- Is this truss statically determinate internally?

- ## 4.
- Truss is statically determinate internally if $m + 3 = 2j$
 $m = 2j - 3$ m is number of members, and j is number of joints in truss



Internal Redundancy or Degree of Static Indeterminacy

Extra Members than required $\rightarrow$ Internal Redundancy

Equilibrium of each joint can be specified by two scalar force equations $\rightarrow$
 $2j$ equations for a truss with “ j ” number of joints
 $\rightarrow$ Known Quantities

For a truss with “ m ” number of two force members, and maximum 3 unknown support reactions $\rightarrow$ Total Unknowns = $m + 3$
 (“ m ” member forces and 3 reactions for externally determinate truss)

Therefore:

$m + 3 = 2j \rightarrow$ Statically Determinate Internally

$m + 3 > 2j \rightarrow$ Statically Indeterminate Internally

$m + 3 < 2j \rightarrow$ Unstable Truss

A necessary condition for Stability but not a sufficient condition since one or more members can be arranged in such a way as not to contribute to stable configuration of the entire truss

Why to Provide Redundant Members?

- To maintain alignment of two members during construction
- To increase stability during construction
- To maintain stability during loading (Ex: to prevent buckling of compression members)
- To provide support if the applied loading is changed
- To act as backup members in case some members fail or require strengthening
- Analysis is difficult but possible

Plane Truss :: Determinacy

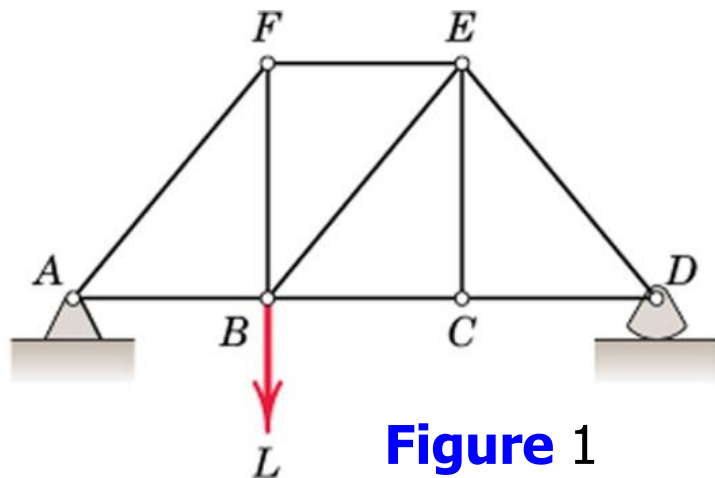


Figure 1

No. of unknown reactions = **3**

No. of equilibrium equations = **3**

: **Statically Determinate (*External*)**

No. of members (m) = 9

No. of joints (j) = 6

No. of unknown reactions (R) = 3

$$\therefore m + R = 2j$$

: **Statically Determinate (*Internal*)**

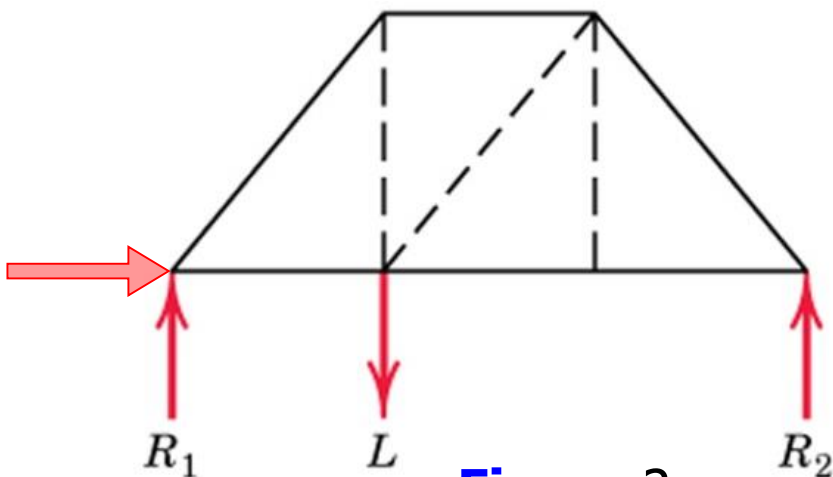
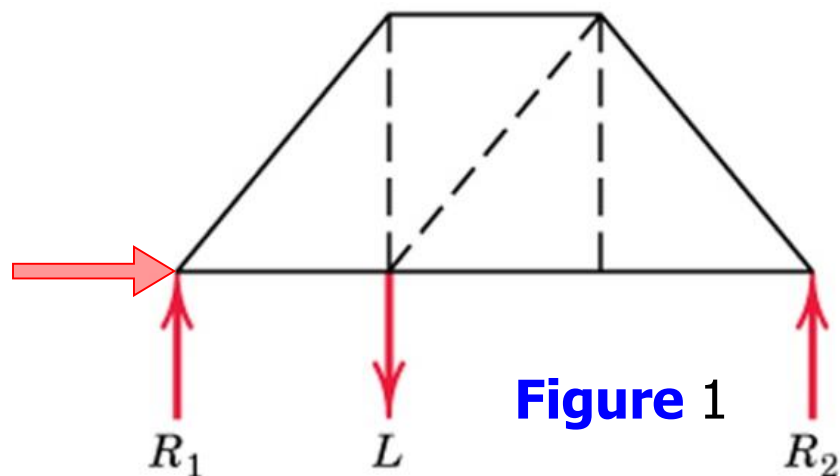
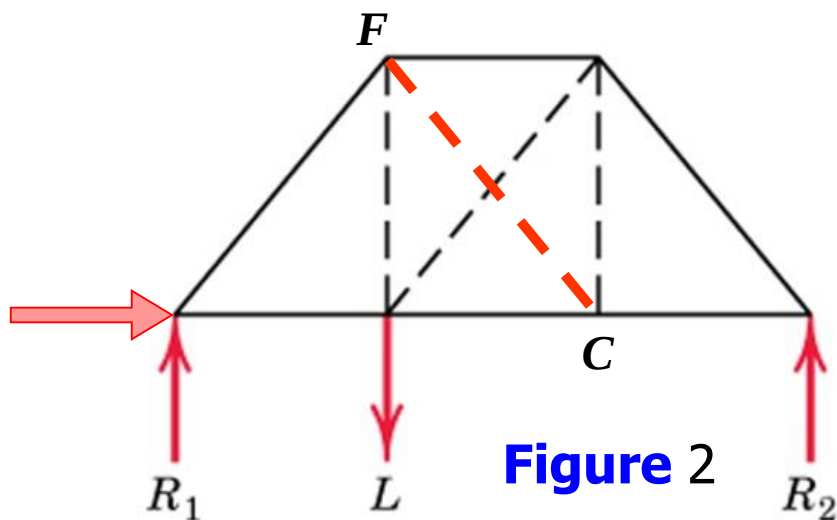


Figure 2

Plane Truss :: Determinacy



- 1. Presence of internal members
 - : Additional sharing for forces
 - : Additional Stability



- 2. Further addition of internal members
 - : Strengthening of Joints C and F
 - : Additional Stability and force sharing
 - : $m + R > 2j$
 - : **Statically Indeterminate (Internal)**

Zero Force Members Conditions

1. If only two non-collinear members form a truss joint and no external load or support reaction is applied to the joint, the two members must be zero force members
2. If three members form a truss joint for which two of the members are collinear, the third member is a zero-force member provided no external force or support reaction is applied to the joint

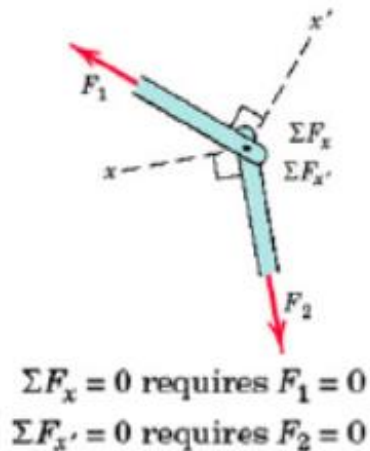


Figure 1

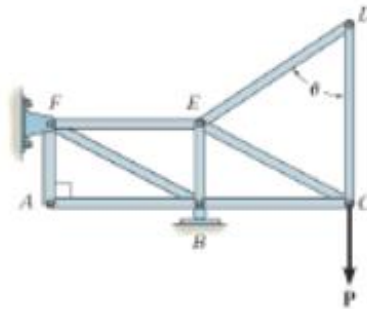


Figure 2

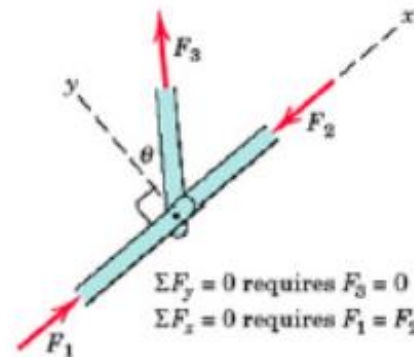


Figure 3

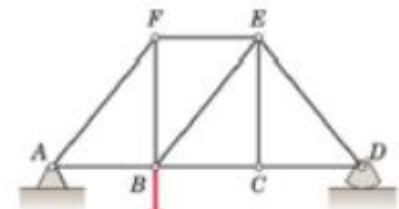


Figure 4

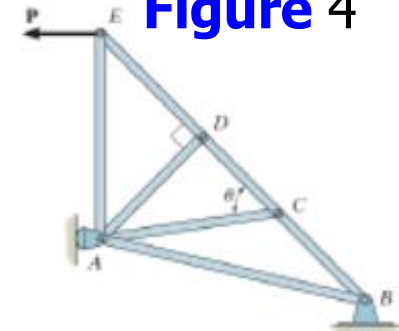
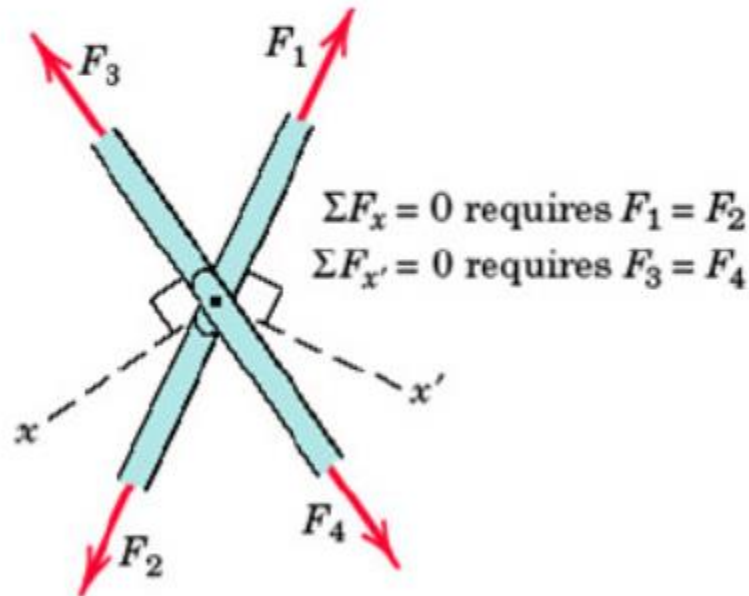


Figure 5

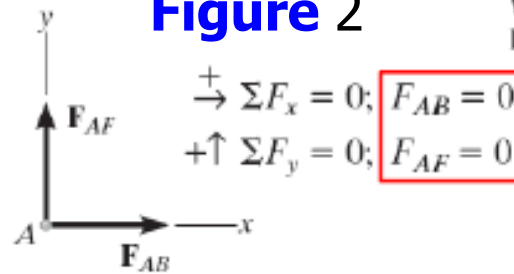
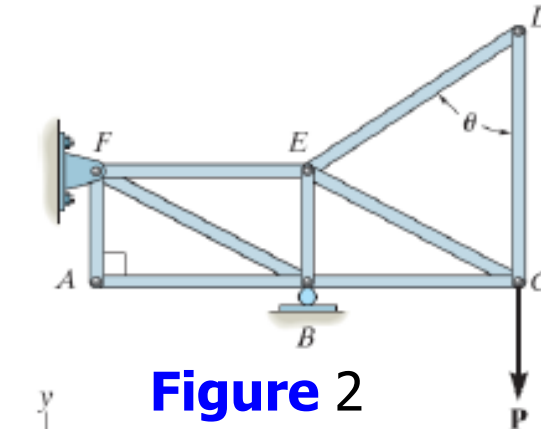
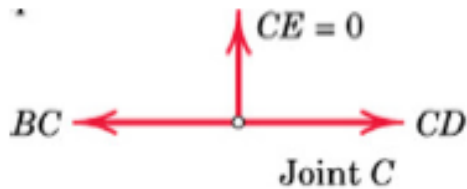
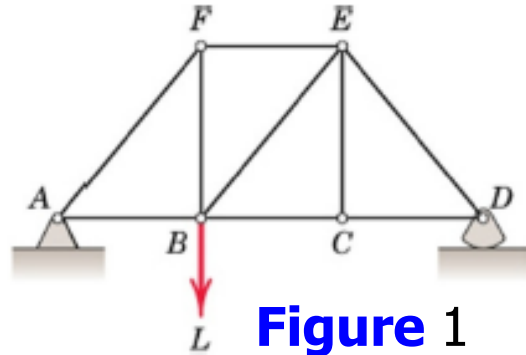
Special Condition

- ❖ When two pairs of collinear members are joined as shown in figure, the forces in each pair must be equal and opposite.

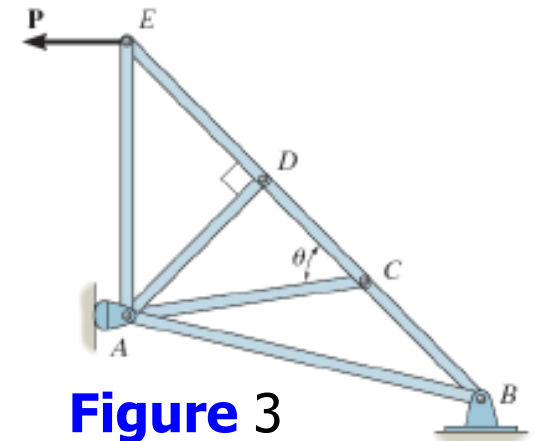


Zero Force Members

Zero Force Members



$+\downarrow \Sigma F_y = 0; F_{DC} \sin \theta = 0; F_{DC} = 0$ since $\sin \theta \neq 0$
 $+\swarrow \Sigma F_x = 0; F_{DE} + 0 = 0; F_{DE} = 0$



$\swarrow \Sigma F_x = 0; F_{DA} = 0$
 $\downarrow \Sigma F_y = 0; F_{DC} = F_{DE}$

$\swarrow \Sigma F_x = 0; F_{CA} \sin \theta = 0$
 $F_{CA} = 0$ since $\sin \theta \neq 0$
 $+\downarrow \Sigma F_y = 0; F_{CB} = F_{CD}$

Figure 1

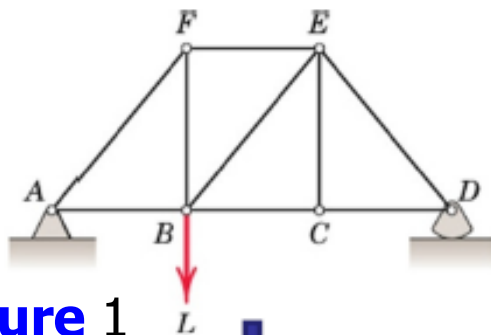


Figure 2

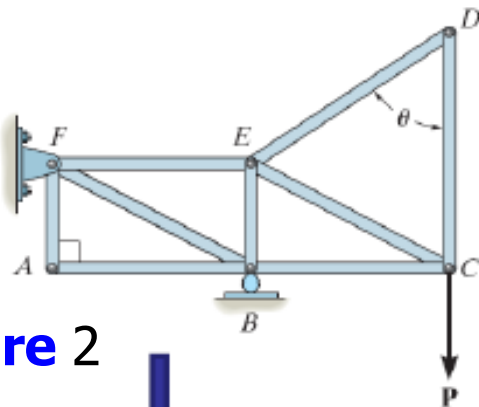
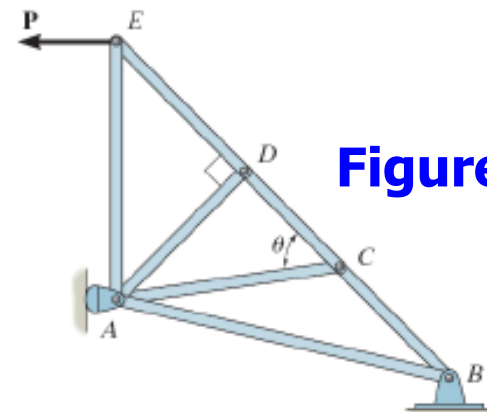


Figure 3



(Problem-2) The truss shown in **Fig.(a)** is loaded by an external force F . Determine the forces at the supports and in the members of the truss shown in **Fig. (b)**?

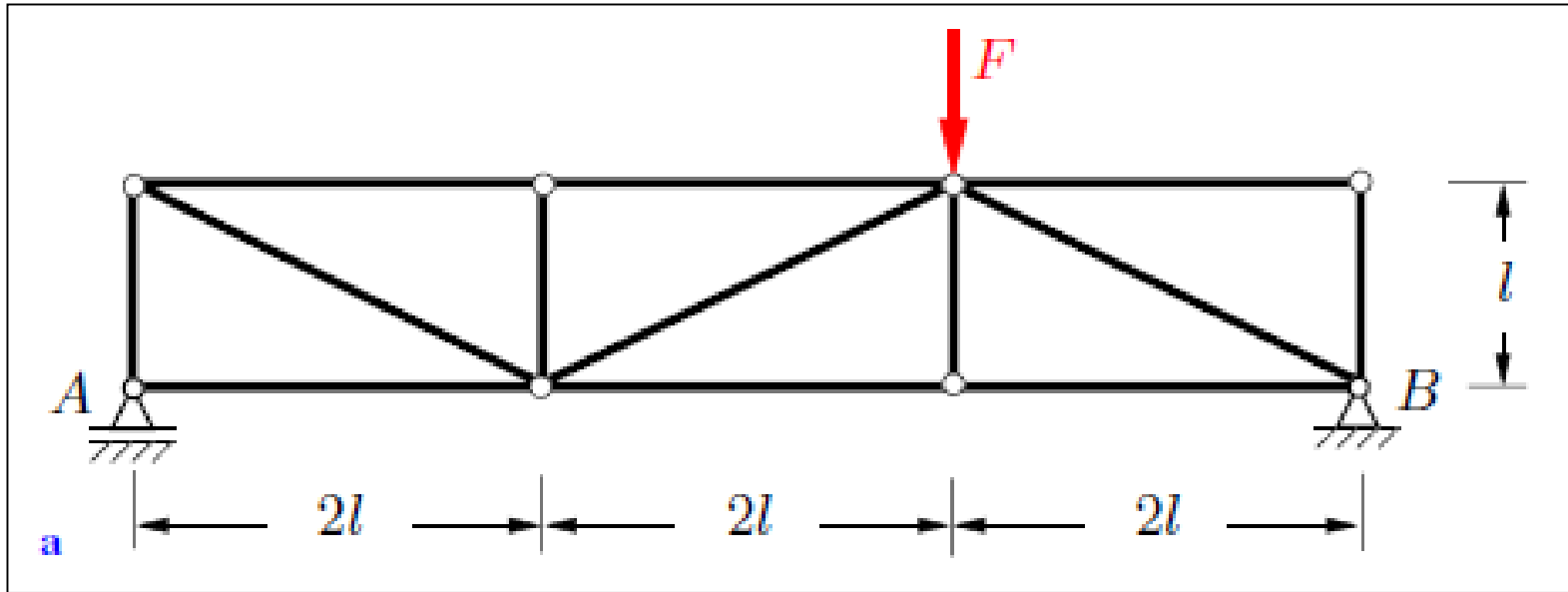
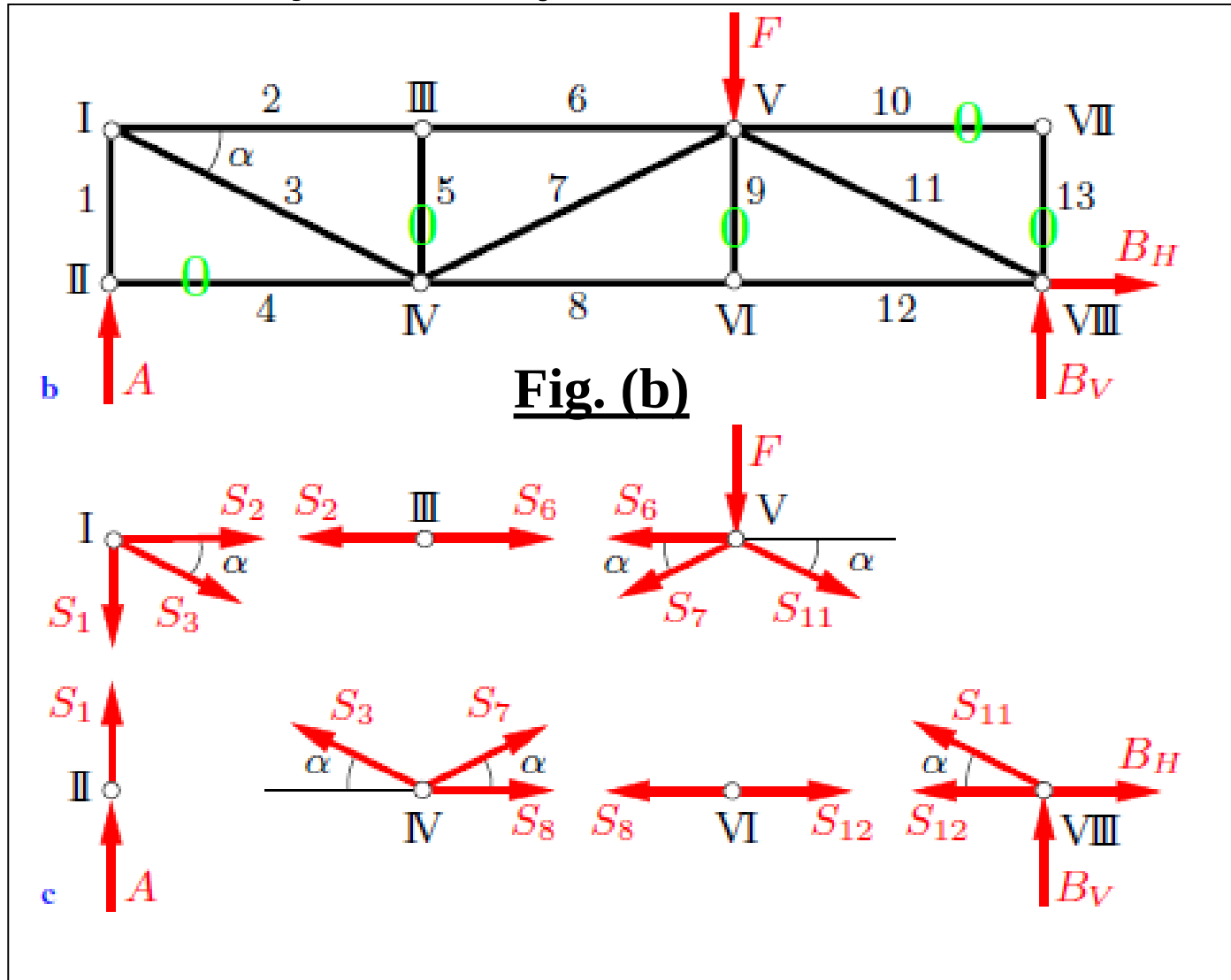
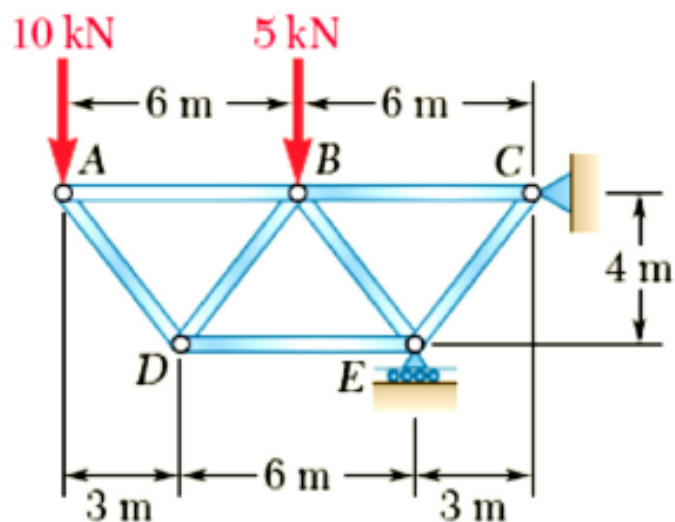


Fig. (a)

- First of all determine the zero force members and eliminate them from the Structure, then finally draw the structure and make two equilibrium equations Of each joint separately by drawing F.B.D's of each joint clearly.



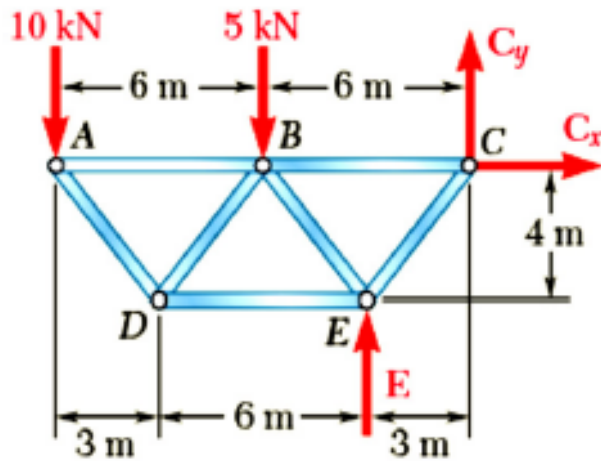


Using the method of joints, determine the force in each member of the truss.

SOLUTION:

- Based on a free-body diagram of the entire truss, solve the 3 equilibrium equations for the reactions at E and C .
- Joint A is subjected to only two unknown member forces. Determine these from the joint equilibrium requirements.
- In succession, determine unknown member forces at joints D , B , and E from joint equilibrium requirements.
- All member forces and support reactions are known at joint C . However, the joint equilibrium requirements may be applied to check the results.

First of all we have to determine the support reactions by applying the three equilibrium equations at C,



SOLUTION:

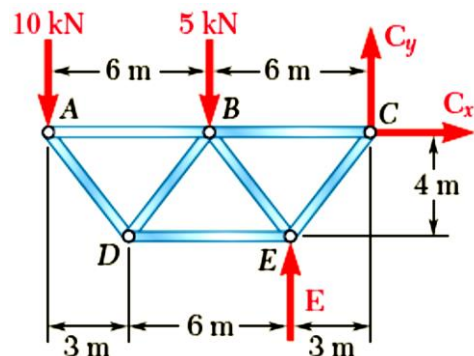
- Based on a free-body diagram of the entire truss, solve the 3 equilibrium equations for the reactions at E and C.

$$\begin{aligned}\sum M_c &= 0 & \text{(i)} \\ &= (10 \text{ kN})(12 \text{ m}) + (5 \text{ kN})(6 \text{ m}) - E(3 \text{ m}) \\ &E = 50 \text{ kN} \uparrow\end{aligned}$$

$$\sum F_x = 0 = C_x \quad \text{(ii)} \quad C_x = 0$$

$$\begin{aligned}\sum F_y &= 0 = -10 \text{ kN} - 5 \text{ kN} + 50 \text{ kN} + C_y & \text{(iii)} \\ &C_y = 35 \text{ kN} \downarrow\end{aligned}$$

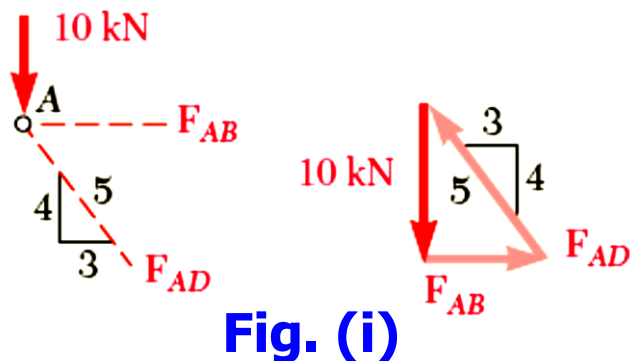
Sample Problem



1. We now solve the problem by moving sequentially from joint to joint and solving the associated FBD for the unknown forces.

❖ Which joint should you start with, and why?

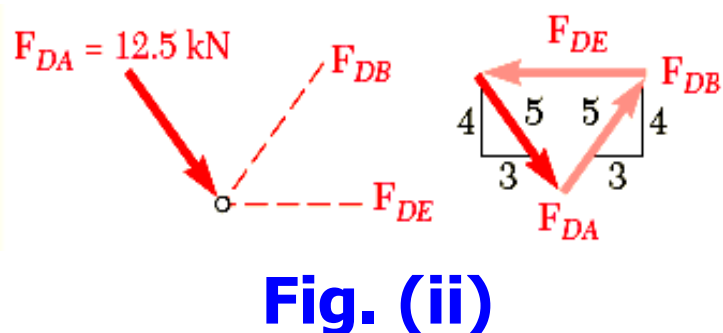
2. Joints A or C are equally good because each has only 2 unknown forces. Use joint A and draw its FBD and find the unknown forces.



$$\frac{10 \text{ kN}}{4} = \frac{F_{AB}}{3} = \frac{F_{AD}}{5}$$

$$\begin{aligned} F_{AB} &= 7.5 \text{ kN } T \\ F_{AD} &= 12.5 \text{ kN } C \end{aligned}$$

❖ Which joint should you move to next, and why?

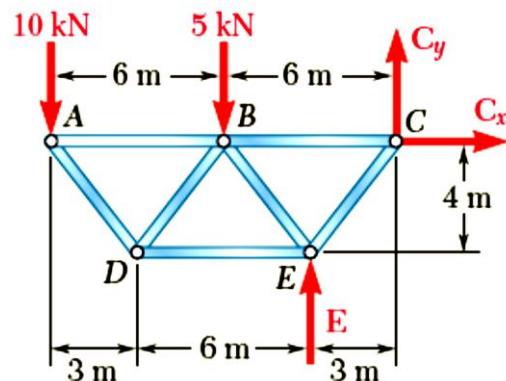


3. Joint D, since it has 2 unknowns remaining (joint B has 3). Draw the FBD and solve.

$$\begin{aligned} F_{DB} &= F_{DA} \\ F_{DE} &= 2\left(\frac{3}{5}\right)F_{DA} \end{aligned}$$

$$\begin{aligned} F_{DB} &= 12.5 \text{ kN } T \\ F_{DE} &= 15 \text{ kN } C \end{aligned}$$

Sample Problem



- There are now only two unknown member forces at joint B. Assume both are in tension.

$$\sum F_y = 0 = -5 \text{ kN} - \frac{4}{5}(12.5 \text{ kN}) - \frac{4}{5} F_{BE} \quad (\text{iv})$$

$$F_{BE} = -18.75 \text{ kN}$$

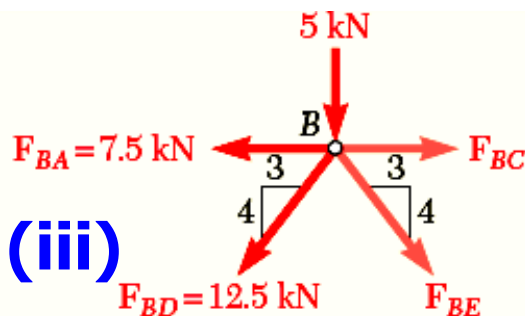
$$F_{BE} = 18.75 \text{ kN } C$$

$$\sum F_x = 0 = F_{BC} - 7.5 \text{ kN} - \frac{3}{5}(12.5 \text{ kN}) - \frac{3}{5}(18.75 \text{ kN}) \quad (\text{v})$$

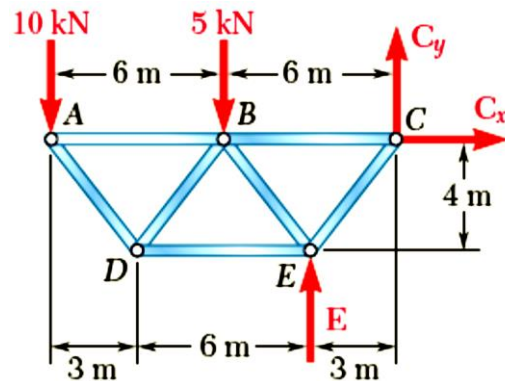
$$F_{BC} = +26.25 \text{ kN}$$

$$F_{BC} = 26.25 \text{ kN } T$$

Fig. (iii)



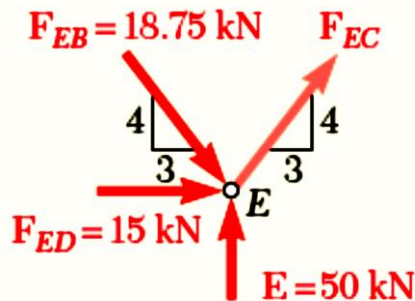
Sample Problem



- There is one remaining unknown member force at joint E (or C). Use joint E and assume the member is in tension.

$$\sum F_x = 0 = \frac{3}{5} F_{EC} + 15 \text{ kN} + \frac{3}{5} (18.75 \text{ kN}) \quad (\text{vi})$$

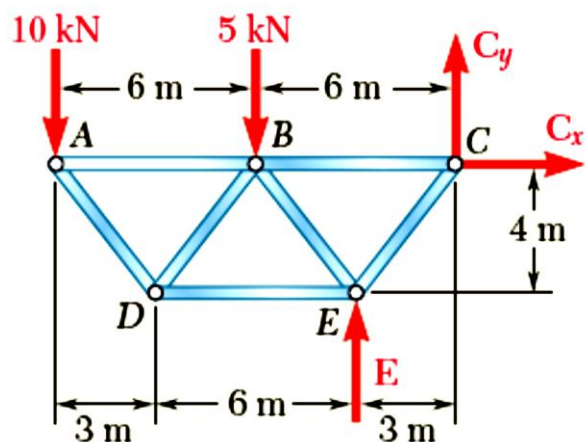
$$F_{EC} = -43.75 \text{ kN}$$



$$F_{EC} = 43.75 \text{ kN } C$$

Fig. (iv)

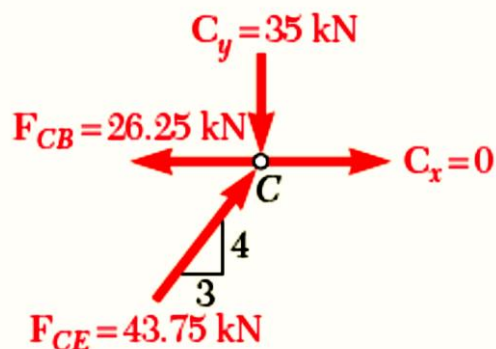
Sample Problem



- All member forces and support reactions are known at joint C. However, the joint equilibrium requirements may be applied to check the results.

$$\sum F_x = -26.25 \text{ kN} + \frac{3}{5}(43.75) \text{ kN} = 0 \quad (\text{checks})$$

$$\sum F_y = -35 \text{ kN} + \frac{4}{5}(43.75) \text{ kN} = 0 \quad (\text{checks})$$



End